

Sandbox (moderate)

```
khadejaq@DESKTOP-3E2P095 x + v
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printk.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1019 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sandbox $((1<<SYS_open)) - cat README
syntax
$ sandbox $((1<<SYS_open)) - cat README
syntax
$ sandbox 32768 - cat README
exec sandbox failed
$ sandbox 32768 - cat README
cat: cannot open README
$ |
```

Sandbox(easy)

```
khadejaq@DESKTOP-3E2P095 x + v
fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printk.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1019 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix Version 6 (v6). xv6 loosely follows the structure and style of v6, xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer (kaashoek,rtm@mit.edu)). The main purpose of xv6 is as a teaching
$ sandbox 32768 README grep xv6 x
grep: cannot open x
$ |
```

Attack(moderate)

```
khadejaq@DESKTOP-3E2P096: ~/xv6-labs-2025$ git push -u origin problemSet2
Username for 'https://github.com': khadeja-q
Password for 'https://github.com':
fatal: unable to access 'https://github.com/khadeja-qureshi/xv6-labs.git/': Error in the HTTP2 framing layer
khadejaq@DESKTOP-3E2P096:~/xv6-labs-2025$ git push -u origin problemSet2
Username for 'https://github.com': khadeja-q
Password for 'https://github.com':
Enumerating objects: 57, done.
Counting objects: 100% (57/57), done.
Delta compression using up to 12 threads
Compressing objects: 100% (29/29), done.
Writing objects: 100% (43/43), 16.37 KiB | 3.27 MiB/s, done.
Total 43 (delta 25), reused 23 (delta 10), pack-reused 0
remote: Resolving deltas: 100% (25/25), completed with 12 local objects.
remote:
remote: Create a pull request for 'problemSet2' on GitHub by visiting:
remote:   https://github.com/khadeja-qureshi/xv6-labs/pull/new/problemSet2
remote:
To https://github.com/khadeja-qureshi/xv6-labs.git
 * [new branch]      problemSet2 -> problemSet2
Branch 'problemSet2' set up to track remote branch 'problemSet2' from 'origin'.
khadejaq@DESKTOP-3E2P096:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/exec.sh user/cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_attack user/_se
cret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1064 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ secret MySecret123
$ attack
MySecret123
$ |
```

MakeGrade

```
khadejaq@DESKTOP-3E2P096: ~/xv6-labs-2025$ make grade
nel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kerne
l/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o
kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/khadejaq/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (3.4s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.6s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.3s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.6s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.1s)
== Test attack ==
$ make qemu-gdb
attack: OK (2.6s)
== Test time ==
time: OK
Score: 45/45
khadejaq@DESKTOP-3E2P096:~/xv6-labs-2025$
```